Question	Answer	Mark	Mark scheme	Additional guidance
21	3 : 4	P1	starts process eg $\overrightarrow{AB} = \mathbf{b} - \mathbf{a}$ oe	
		P1	for process to find $\overrightarrow{OM} = \mathbf{a} + \frac{1}{2}$ "($\mathbf{b} - \mathbf{a}$)"oe ($= \frac{1}{2}(\mathbf{a} + \mathbf{b})$)	
		P1	for process to find $\overrightarrow{AP} = -\mathbf{a} + \frac{3}{5}$ "($\frac{1}{2}\mathbf{a} + \frac{1}{2}\mathbf{b}$)" oe **or** (indep) for $\overrightarrow{AN} = -\mathbf{a} +$ "k"$\mathbf{b}$	
		P1	process to find "k" using $\overrightarrow{AN} = -\mathbf{a} +$ "k"$\mathbf{b}$ as a multiple of $\overrightarrow{AP}$	
		A1	cao	
			ALTERNATIVE	
		P1	for producing OM to C such that AC is parallel to OB	Formal geometric reasoning relating to congruent and similar triangles is not required
		P1	for process to show that $MC = OM$, using congruent triangles ACM and BOM for process to find PC as a multiple of OM/5 (= 7OM/5)	
		P1	for process to find ON as a multiple of AC(OB) (= 3OB/7) using similar	
		P1	triangles ACP and NOP	
		A1	cao	